**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 161 · Paper 1: Calculations and Extended Answers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a constant?

Q2. What does an IDE's run-time environment let you do?
A. Run and test code
B. Repair physical CPU faults
C. Rearrange HDD fragments
D. Manage network user accounts

Q3. Where are the MSB and LSB in an eight-bit unsigned number?

Q4. A file stores text, not a photo of a page. “Every letter A is stored as a tiny bitmap.” Correct this. How is a letter stored and read?

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

Show formulas — Recall and write the calculation rule before substitutions.

Keep units consistent — Convert bits/bytes and MB/GB before arithmetic.

Answer command words — Describe what happens, explain why and evaluate trade-offs.

Earn full tariff — Add linked causal explanations, not repeated versions of one fact.

### New learning

Show formulas: Recall and write the calculation rule before substitutions.

Keep units consistent: Convert bits/bytes and MB/GB before arithmetic.

Answer command words: Describe what happens, explain why and evaluate trade-offs.

Earn full tariff: Add linked causal explanations, not repeated versions of one fact.

### Worked example

Calculate image size: 120×80 pixels at 4-bit colour depth, in bytes.

1. Pixels: 120×80 = 9600.

2. Size: 9600×4=38,400 bits.

3. Divide eight = 4800 bytes.

4. Identify all figures and correct units.

## We Do · Practise with guidance

### Guided 1

Calculate sound file bits: 8000 Hz, 5 seconds, 8 bits/sample (mono).

Worked method: No header/compression assumptions.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Change the bit depth and predict the new uncompressed raw data size in bytes.

```
width = 12
height = 10
bitsPerPixel = 4
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

write

### Question 1

Calculate 2560 bits in bytes. [2 marks]

### Question 2

Convert 3 MB to kB under decimal units. [2 marks]

### Question 3

Explain why two distinct SSD benefits may score better than repeating 'fast'. [3 marks]

## Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

When calculating sound size, sampling rate can be ignored.





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A named value that stays the same while the program runs.

Q2. A — Run and test code

Q3. The MSB is the leftmost bit, weight 128; the LSB is the rightmost bit, weight 1.

Q4. Each character has a number code stored in binary. Both computers need a matching encoding to show the intended characters.

### I Do / We Do pages · Guided practice

Q1. No header/compression assumptions.

### You Do pages · Code First

```
width = 12
height = 10
bitsPerPixel = 8
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

Expected result: 120

### You Do pages · Independent questions

Q1. 2560 ÷ 8 = 320 bytes.

Q2. 3000 kB.

Q3. Specific benefits (e.g. impact durability and access time) address different marking points.

### Exit Ticket · Model answer

False

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**